

Profile:

Robotics software developer with hands-on experience deploying HRI systems on a physical TIAGo robot, competed at RoboCup@Home 2026. Experienced in building end-to-end technical projects spanning machine learning, SaaS systems, and ROS-based robotics, within Linux environments, with a growing interest in robotic automation and applied AI.

Experience:

- **Robotics Software Developer & Development Lead | Learning Autonomous Service Robots (LASR) Team @ KCL (October 2025 - Current)**
 - Helped manage a 10-person engineering team for the RoboCup@Home league, coordinating task deadlines and code-architecture onboarding for new members, while outputting key components for the codebase.
 - Architected HRI software pipelines (person-following, seating navigation) and led hardware testing/deployment for General Purpose Service Robot (GPSR) tasks.
 - Represented KCL at RoboCup 2026 in Incheon, South Korea, presenting posters on the team's open-source codebase to international robotics researchers, and supporting the development of 2 key deliverable products.
- **Graduate Teaching Assistant | King's College London (October 2026 - Ongoing):**
 - Facilitate hands-on robotics labs for Level 5 modules, instructing students on Python software deployment using ROS and Linux.
 - Provide technical support and troubleshooting for student hardware experiments on TurtleBot3 Waffle Pi systems.
- **Sales Assistant | Screwfix (September 2024 - Ongoing):**
 - Built rapport with regular customers by identifying patterns in issues and applying proven solutions, demonstrating problem-solving and customer focus.
 - Supported warehouse operations by organising inventory, restocking efficiently, and maintaining a clean workspace to improve productivity.
- **King's Undergraduate Research Fellow (KURF) | King's College London (June 2026 - July 2026):**
 - Completed a funded research under Dr. Matteo Leonetti on the project "Robots for Healthy Ageing: Engineering Socially Assistive AI".
 - Designed and implemented Human-Robot Interaction (HRI) behaviours for the PAL Robotics TIAGo platform, deploying machine learning algorithms and LLMs for robust voice command recognition.
 - Executed structured research cycles, including literature reviews, experiment planning, and quantitative data analysis on robot success rates and navigation safety for frail populations.

Extracurricular

- **Cadet Staff Sergeant | Combined Cadet Force (November 2019 - July 2024):**
 - Led, instructed, and mentored younger cadets while representing the contingent at recruitment events, showcasing leadership and teamwork. As well as organised and managed cadet activities, ensuring high standards of discipline and performance during drills and exercises.
 - Displayed resilience, adaptability, and clear decision-making under pressure by often commanding training exercises and expeditions, leading a platoon (~24) of cadets on multiple occasions.

Education:

- **King's College London: [2024 - 2027]**
 - Artificial Intelligence BSc
 - Current Grade Classification: 1st
- **St Ignatius College [Sept 2017 - June 2024]**
 - 3 A-Levels + EPQ : Achieved A*, A, A in Maths, Further Maths and Computer Science respectively.
 - 9 GCSE (Attained a grade average of 8.2)

Projects:

- Loan Approval Predictor (Logistic Regression / Python)
 - Designed and implemented a logistic regression neural network from scratch in Python to predict mortgage approval outcomes.
 - Focused on model architecture, numerical optimisation, and data preprocessing without high-level ML frameworks.
- Music Recommendation SaaS (Full-Stack / ML / DevOps / Git / Cloud)
 - Built a fully containerised and deployable SaaS platform integrating computer vision and ML models.
 - Used YOLO and BitNet to infer user context and generate real-time music recommendations.
 - Emphasised modular design, containerisation, and deployment-ready architecture.
- LASR Open-Source Codebase & HRI Software (ROS1/ROS2 / Python / Open-Source)
 - Researched and ported the official LASR team codebase to newer software architectures, successfully migrating systems from ROS 1 to ROS 2.
 - Developed and integrated Human-Robot Interaction (HRI) behaviours such as computer vision based guest-seating algorithms and object handover and placement on a physical TIAGo robot.
 - Tested and deployed a 'natural language instructions to executable actions' pipeline, adding new skills for the robot to execute, enabling it to autonomously plan and execute tasks of varying complexity from diverse inputs.
 - Produced competition-ready code deployed internationally at RoboCup, and successfully demonstrated the software in public settings including the KCL Informatics Showcase and Open Days.